

Lunar deposits of possible pyroclastic origin

GRANT H. HEIKEN, DAVID S. McKAY

NASA-Johnson Space Center, Planetary and Earth Sciences Division,
Houston, Texas 77058, U.S.A.

and

R. W. BROWN

Lockheed Electronics Company, Houston, Texas 77058, U.S.A.

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Abstract—Glass droplets of possible pyroclastic origin are present in the lunar regolith at the Apollo 11, 15, and 17 sites. The droplets may be derived from deposits, interbedded with mare lava flows, which have been partly mixed into the regolith by impact processes. Orange glass droplets from the Apollo 17 site (spheres, ovoids, broken droplets) are both chemically and texturally homogeneous and have rare olivine phenocrysts. None of the droplets contain shock damaged crystals which are common in glass produced during meteorite impacts. The droplets are similar to those formed in terrestrial lava fountains and are here interpreted as tephra.

The homogeneous glass droplets sampled at the Apollo 11, 15 and 17 sites are located on or close to mare basin rims. Vents for the youngest mare lava flows, located near basin rims, have been identified photogeologically. Dark mantle deposits, interpreted as pyroclastic blankets in some locations, and numerous rilles are also present on the mare surface, near basin rims. The glass droplets, having ages nearly contemporaneous with the associated mare lavas, may be concentrated locally near such vent areas. This association is in accordance with the limited extent of ash deposits from terrestrial lava fountains (≤ 10 km from the vent).

INTRODUCTION

SINCE the first lunar samples were returned to Earth in 1969 it has been clear that the Moon has had an extensive early history of volcanic activity. The primary product of that volcanic activity was the mare basalts which cover all the mare basins on the nearside of the Moon and which occur in deep basins on the lunar farside. However, no samples of obvious pyroclastic origin, such as volcanic ash or pumice, were reported by the early Apollo missions.

The dark mantle deposits of the Moon have been interpreted as being possible pyroclastic deposits (CARR, 1966; SCHMITT *et al.*, 1967; SALISBURY *et al.*, 1968; EL BAZ, 1972; HEAD *et al.*, 1972; McGETCHIN and HEAD, 1973; LUCCHITTA, 1973). These deposits, identified through Earth-based telescopes and orbital photography, consist of continuous to patchy smooth surfaces with a low albedo. Good examples of dark mantle deposits are located along the rim of Mare Serenitatis, and in the floors of craters Alphonsus and Humboldt; some of the dark mantle deposits are associated with orange and red deposits, seen by Apollo crews, which are exposed in crater walls. In many cases, the dark mantle is associated with low rimmed craters particularly along the east rim of Mare Serenitatis (Fig. 1). The morphology of these craters is similar to the broad, low profile shape predicted by McGETCHIN and HEAD (1973) for lunar cinder cones based on studies of cinder cone formation at Etna and Stromboli, Italy. In the Valley of Taurus-Littrow, the Apollo 17 site, the dark mantle was interpreted before the mission as pyroclastic ejecta which draped mare lavas (SCOTT *et al.*, 1973; EL BAZ, 1973).

One of the primary objectives of the last Apollo mission, Apollo 17, was to sample these smooth, dark mantle deposits in hopes that they would be relatively young volcanic deposits.

SAMPLES FROM THE DARK MANTLE

A. *Normal soils*

At the Apollo 17 site, 24 soil samples were collected from the dark mantle; selected size fractions of 10 of these soil samples have been studied petrographically. The 90–150 μm size fraction of these soils consists of 27–66 per cent comminuted basalt (basalt fragments and clinopyroxene, plagioclase and ilmenite grains), 5–7 per cent breccia fragments, 9–56 per cent agglutinates (mineral and lithic detritus bonded together with clusters of nearly opaque brown glass), and 2–18 per cent orange and black spheres (APOLLO 17 P.E.T., 1973, BUTLER, 1973; HEIKEN and MCKAY, 1973). Other size fractions of these soils contain the same components but relative proportions vary slightly.

B. *Orange and black soils*

At the west end of the valley traversed by the Apollo 17 crew, the regolith and mare lavas are overlain by a relatively thin, bright landslide deposit derived from the South Massif. This deposit is penetrated by Shorty Crater, a 120 m diameter, fresh crater with a dark rim (Fig. 2). The walls of Shorty Crater are smeared with alternating bands of (1) basalt blocks and (2) dark gray, smooth, unconsolidated material containing faint orange bands.

It is on the rim of Shorty Crater that several unique 'soils' were collected. Orange 'soil' forms a 0.8 m wide band with sharp boundaries between two gray soils; underlying the orange 'soil' at a depth of about 25–30 cm is a fine-grained black 'soil' (SCHMITT, 1973). The orange band consists of a weakly coherent aggregate consisting of 100 per cent orange glass spheres and sphere fragments. Samples from the orange band consisted of loose material and cohesive aggregates ('clods') up to 6 cm in diameter. The largest sample exhibited a distinct banding, with a reddish brown, 1–2 cm thick, outer band and a reddish to pinkish gray core. The nature of this banding has not been determined.

In addition to the scoop sample of the orange band, a 70 cm long core tube was driven into the band. Dust clinging to the exterior of this core tube shows an abrupt change in orange material to black material at a depth of about 25 cm (SCHMITT, 1973). Material from the lower end of this core consists primarily of black, partly crystallized glass droplets (APOLLO 17 P.E.T., 1973).

CHARACTERISTICS OF THE ORANGE AND BLACK GLASS DROPLETS

A. *General description*

The black and orange 'soils' both have a mean grain size of about 40 μm , with droplets ranging from less than 1 μm to about a millimeter in diameter and are exceptionally well sorted for lunar soils. Grain size distributions for the orange and black droplets (74001 and 74220) are nearly identical (Fig. 3) and are distinctive from Apollo 17 'dark mantle' soils.

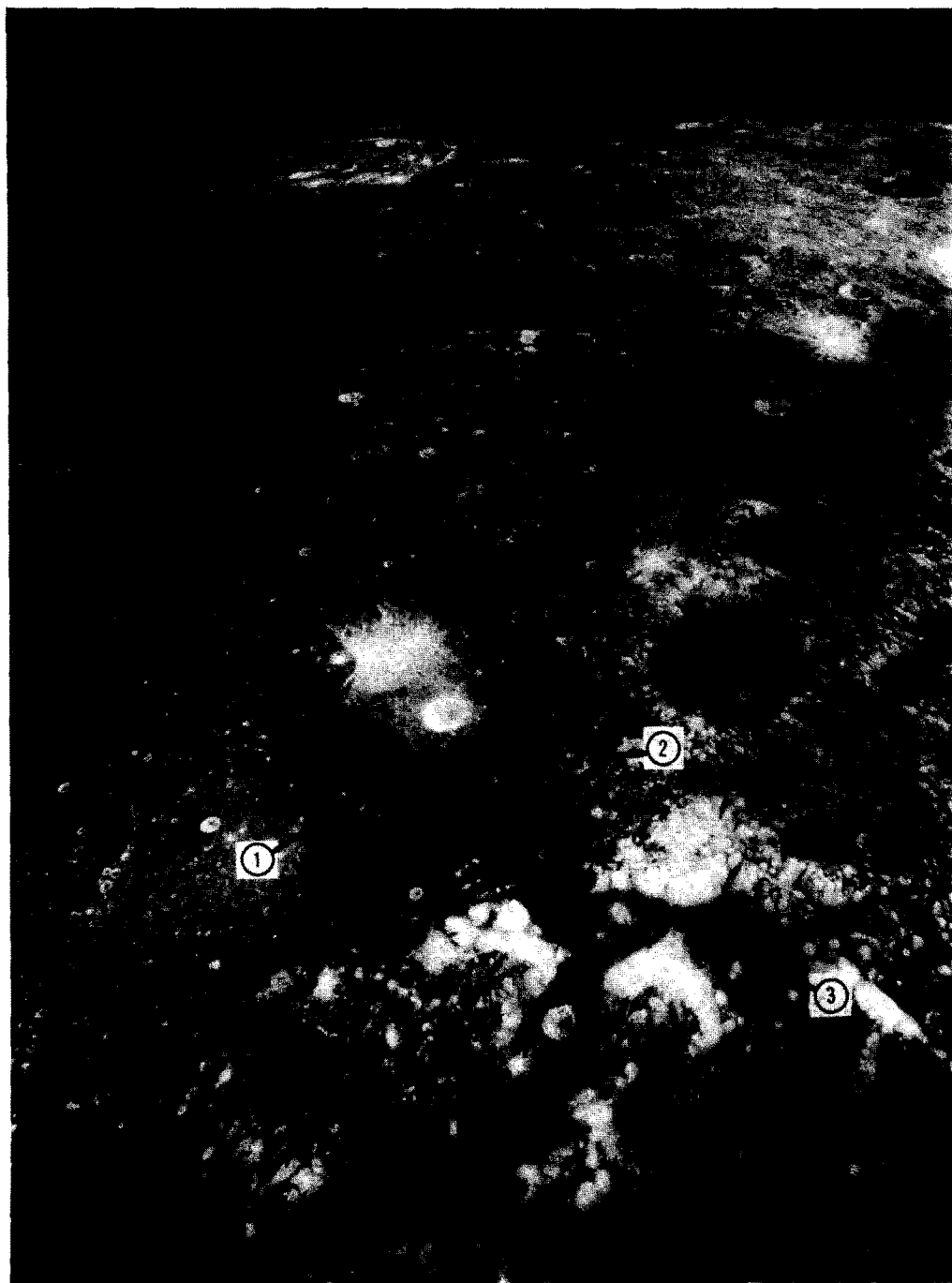


Fig. 1. Oblique, north facing view of the east edge of Mare Serenitatis. (1) Contact between the dark mantle deposits and higher albedo plains material of Mare Serenitatis. (2) Black haloed craters, located in the sculptured highland terrain around the rim of the basin. (3) Valley of Taurus-Littrow, the Apollo 17 landing site. Low albedo material on the valley floor is the dark mantle discussed in this report. The valley is 11.5 km wide at the east end (right side of photo).
Apollo 15 Metric photograph 1519.



Fig. 2. Panorama and sketch map of Shorty Crater. The eastern—most extent of EVA no. 2 (Extra vehicular activity) reached Shorty Crater, located on the valley (mare) floor. The 110 m diameter crater penetrates the 'light mantle' a thin avalanche deposit overlying the mare basalt and regolith. The crater is one of the freshest (youngest) visited by the crew of Apollo 17. Explanation: Solid line = crater rim, dashed line = contacts between photogeologic units, alternate dot-dashed line = edge of crater floor. Photogeologic units: A. Dark grey 'smooth' deposits, containing $\approx 10-20$ per cent blocks >25 cm. The percentage of blocks is a visual estimate. Some sections of A exhibit a faint orange hue. This unit *may* have been sampled in the bottom of core 74001 (the black droplets) and may also contain orange glass. B. Extremely blocky deposits, containing >50 per cent angular medium to light grey blocks. Unit B was sampled in the 5 m long boulder pile immediately behind the Lunar Rover in this photograph. The samples consist of coarse grained olivine basalt (BUTLER, 1973). C. Dark grey deposits with an orange hue. This unit is similar to A, but is darker, finer grained and contains less blocks. D. Orange-grey deposits with a smooth surface. Interpreted as fine grained elastic material similar to sample 74220 (the orange glass), which was sampled in unit D (near the Lunar Rover on the photograph). E. Very dark grey, smooth deposits, with no orange hue. It is suggested here that this unit is similar to 74001 (the black spheres). The lack of an orange hue suggests that orange glass spheres are not present in this unit.

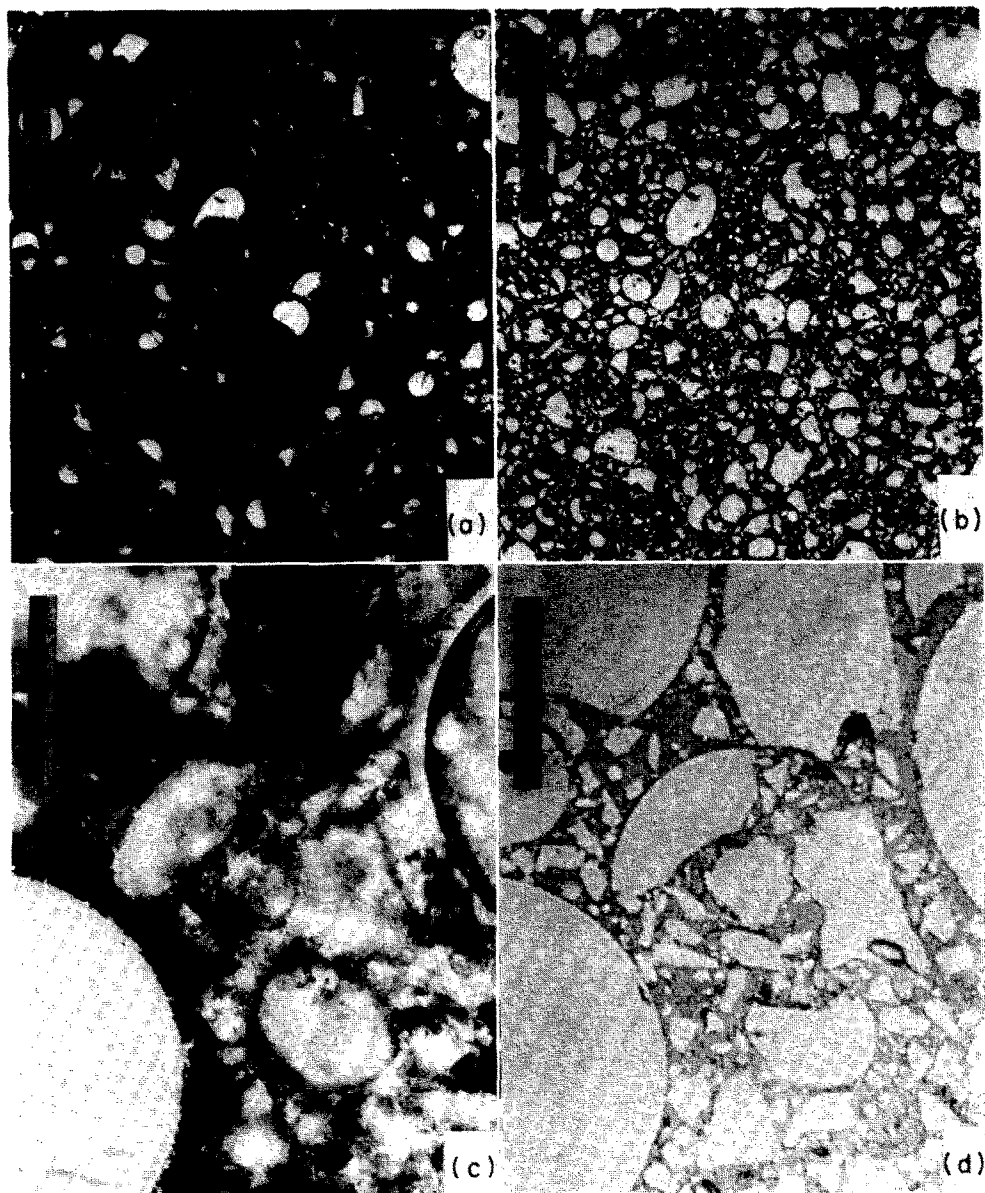


Fig. 4. Sample 74220, friable rock or 'clod' collected at Shorty Crater. The sample consists of 100 per cent orange glass and partly crystallized orange glass. A. Photomicrograph of orange glass droplets and broken droplets. Transmitted light; scale is 0.5 mm. B. Same as A, in reflected light. Scale is 0.5 mm. C. Photomicrograph of matrix from 'clod' of orange droplets. Transmitted light; scale = 50 μ m. D. Same as C, in reflected light. Scale is 50 μ m.

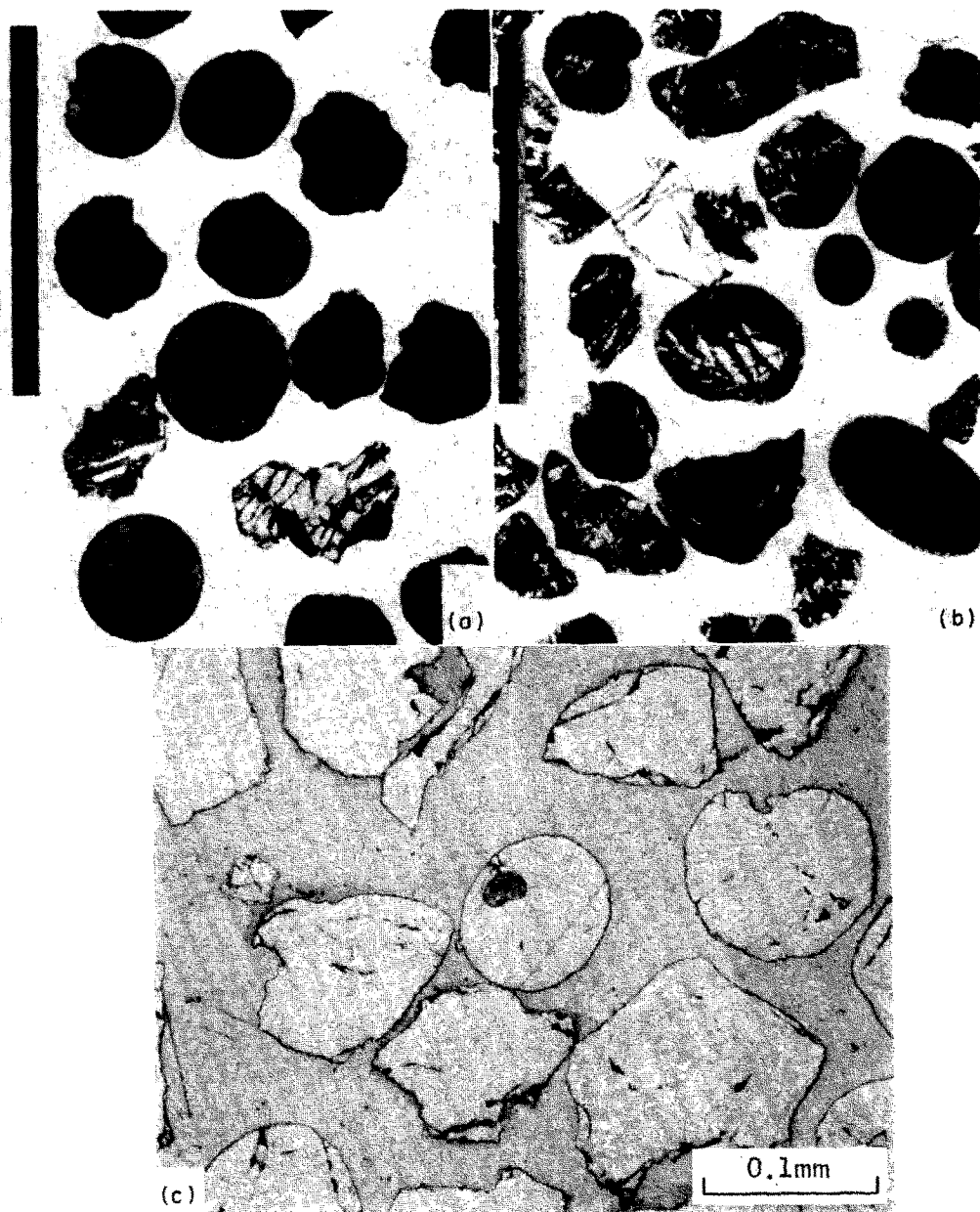


Fig. 5. Thin sections of 'Soil' samples from Station 4, Shorty Crater. A. Photomicrograph of sample 74220,6, mostly orange glass and partly crystallized orange glass droplets. 'Contamination' from the trenching and sampling process is present as minor amounts of basalt and vitric breccia fragments, visible in the lower part of the photograph. Scale is 1 mm. B. Photomicrograph of sample 74001,2, consisting mostly of black droplets exhibiting dendritic textures. Many are broken droplet fragments. Scale is 1 mm. C. Reflected light photomicrograph of 74001,2 (90–150 μm size fraction). Two droplets with vesicles are visible including the droplet toward the upper right part of the photograph which contains a small vesicle open to the outer surface. See also Fig. 7b.

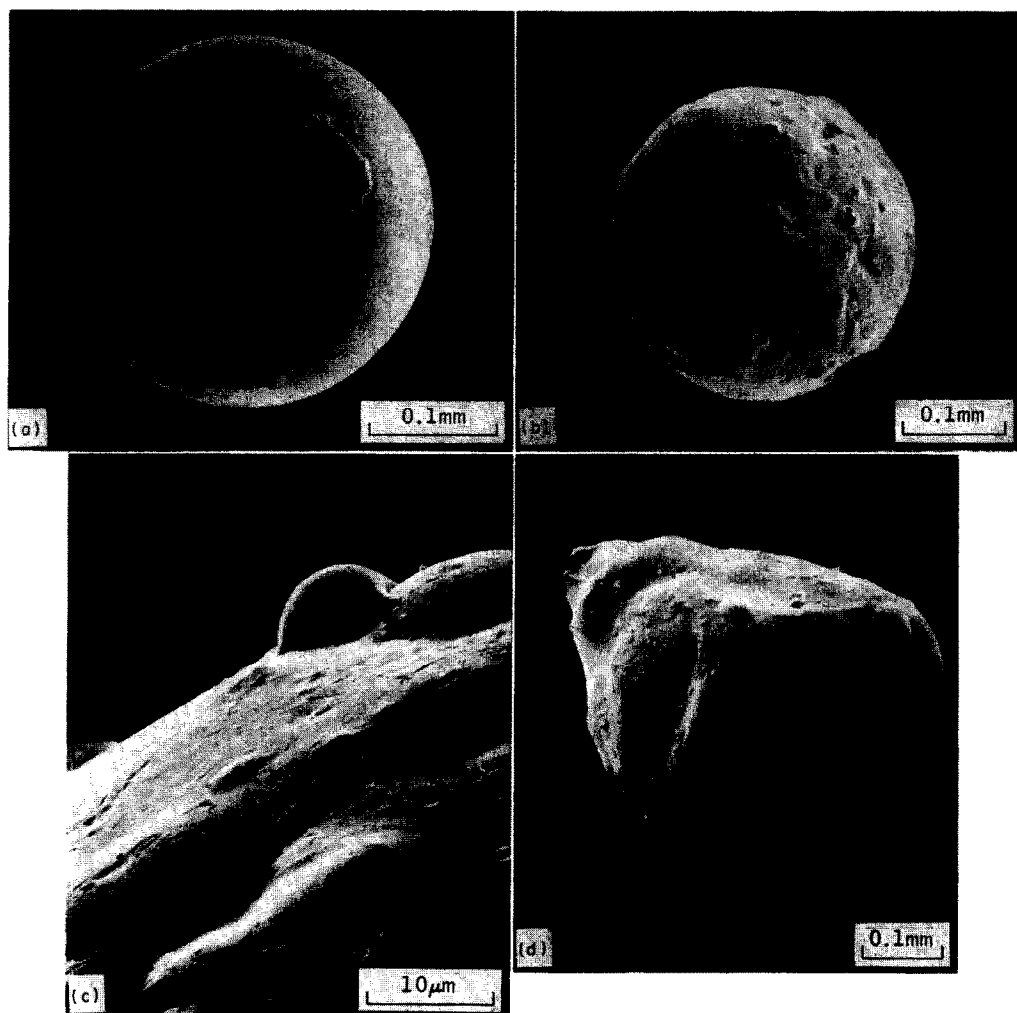


Fig. 6. Scanning electron microscope photographs of droplets from 74220. A. A simple, relatively clean droplet with minor fracturing from low velocity impact. B. A composite droplet which has on its surface multiple adhering droplets of identical composition. C. Close-up of the surface of the droplet in B. The bumpy surface results from multiple adhering droplets which were hot and capable of plastic deformation at the time they collided with the host droplet. The very fine texture results from the presence of micromounds (see Fig. 8A). D. A large olivine phenocryst (Fo_{82}) partially covered with a thin coating of partly crystallized glass.

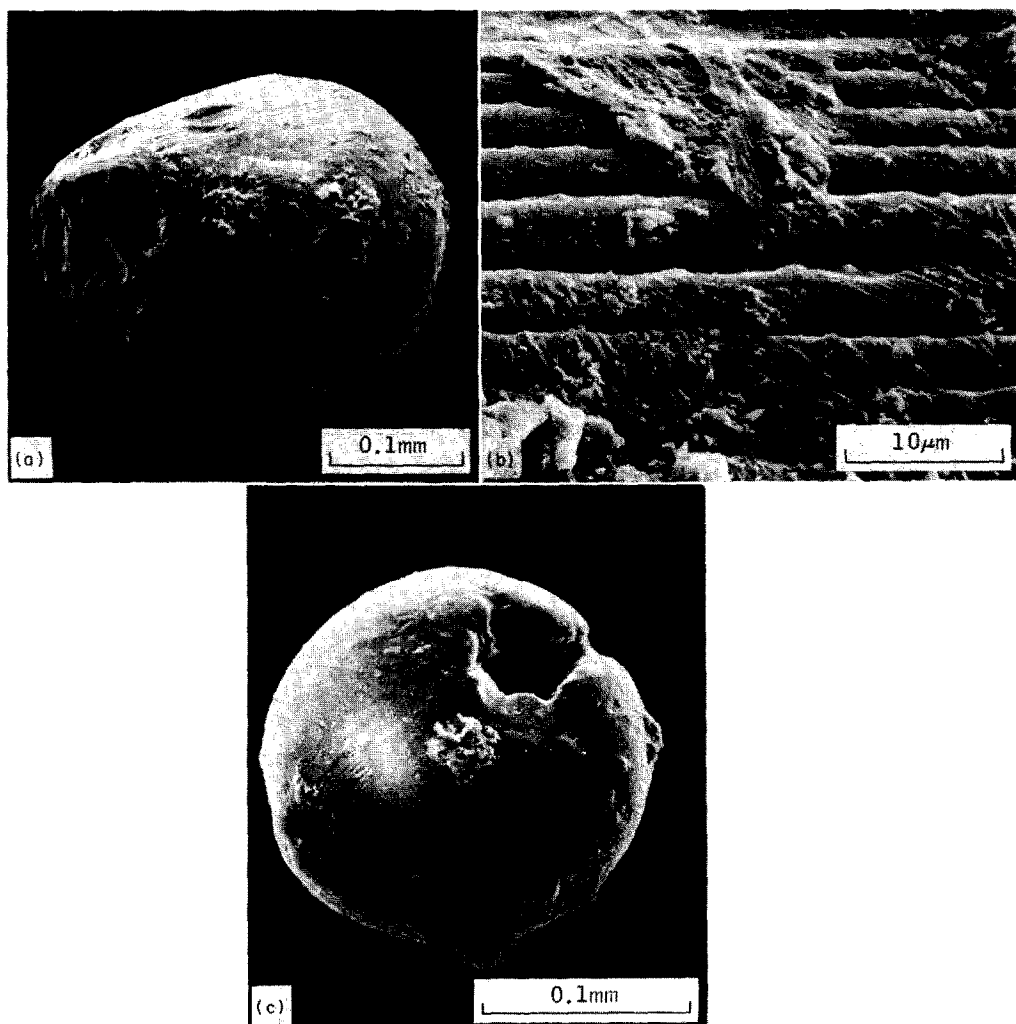


Fig. 7. Scanning electron microscope photographs of droplets from 74001. A. A crystallized barred droplet which displays parallel ilmenite bars intersecting the outer surface. B. Close-up of the droplet in A. showing a thin glass coating partially covering the ilmenite bars. The coating has been scraped by small impacts while still fluid. C. Droplet with vesicle open to the exterior. This droplet is partly crystallized and displays ilmenite bars in a region to the left of center.

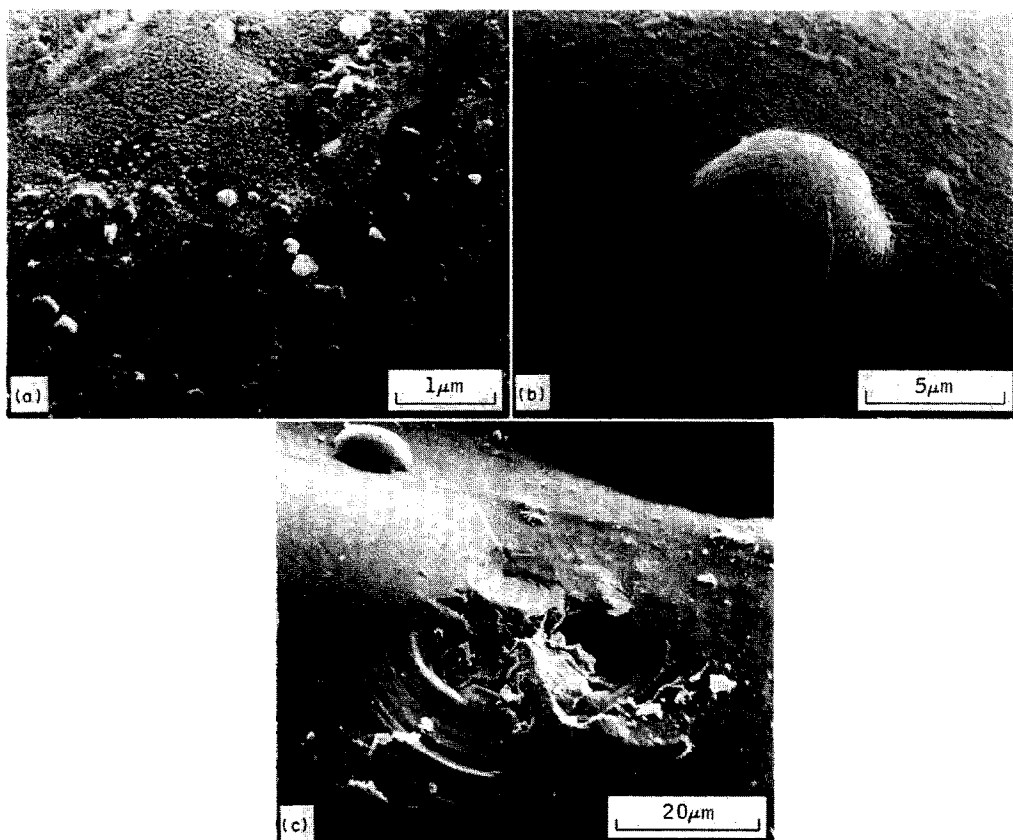


Fig. 8. Scanning electron microscope photographs. A. Close-up of the exterior of a droplet from 74001 showing micromounds ranging in size from less than $200 \text{ \AA}$ to about $500 \text{ \AA}$. These mounds, interpreted as a vapor condensate have been partially disturbed by abrasion and scraping while still hot. B. Close-up of the surface of a 0.1 mm diameter sphere from a Hawaiian fire fountain eruption. The textured surface is interpreted as partially coalesced vapor condensate analogous to the micromounds on the lunar droplets. The large mound is an adhering glass droplet which was attached to the host droplet while still hot and somewhat fluid. C. Impact on a droplet of 74220. This impact has fractured and spalled the host droplet but it is not a micrometeorite impact. It was caused by the relatively low velocity impact of a smaller droplet.

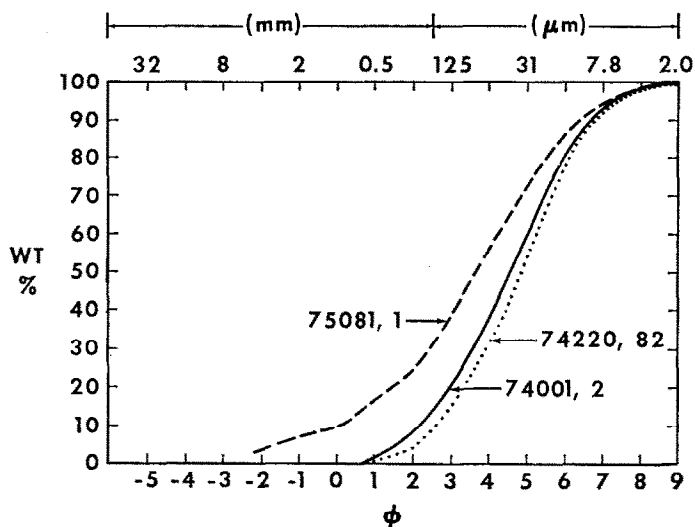


Fig. 3. Grain size distributions for samples 74220,82 and 74001,2. Mean grain sizes are $40\text{ }\mu\text{m}$ for 74220 and $43\text{ }\mu\text{m}$ for 74001. The grain size distributions are nearly identical. For comparison is soil 75081, from the rim of Camelot Crater on the mare surface.

The orange glass from 74220 consists of simple spheres and composite forms (Figs. 4 and 5). The latter are coated with droplets which range in shape from nearly complete spheres to low, hemispherical plates. Analysis with an energy dispersive X-ray spectrometer on the scanning microscope indicates that droplets adhering to the surface are the same composition as the host droplet (McKAY and HEIKEN, 1973; CARTER *et al.*, 1973).

The glass droplets in the orange soil (74220) are petrographically homogeneous and none contain the broken lithic or mineral fragments characteristic of many impact-generated glass droplets (CHAO, 1968; HEIKEN, 1972). The refractive index of the glass is 1.720. About 0.1 per cent of the droplets contain euhedral olivine phenocrysts as large as 0.5 mm with compositions ranging from Fo_{76} to Fo_{82} (McKAY and HEIKEN, 1973; PRINZ *et al.*, 1973; REID *et al.*, 1973a). Nearly half of the orange droplets are partly or completely crystallized; textures range from small sheaf-like bundles of thin ilmenite crystals to chondrule-like forms with parallel intergrowths of ilmenite and olivine which cross the droplets (Fig. 5) (McKAY and HEIKEN, 1973). Most of the partly crystallized droplets are composite, consisting of an ovoid or sphere coated with smaller droplets (Fig. 5a and 6).

The black droplets from 74001 consist of mostly composite forms (Figs. 5c and 7) and are partly to completely crystallized equivalents of orange glass droplets in all sizes (Fig. 3). Ilmenite and olivine crystals which cross the droplets are in some cases visible on the grain surface as parallel ridges, with a thin glass coat (Figs. 7a and b). Thirty-two per cent of the black droplets contain euhedral olivine phenocrysts (Fo_{76} – Fo_{81}). The phenocrysts in the black droplets are more Mg rich than the dendritic olivines (Table 1). The olivine phenocrysts in 74001,2 have the same composition as reported for phenocrysts in the orange glass [74220,66A (250–500 μm

Table 1. Chemical analyses of olivine from 74001,2 (250–500 μm size fraction)

Phenocrysts							
	74001,2-1	74001,2-14	74001,2-8	74001,2-12	74001,2-13		
SiO ₂	39.06	38.84	38.99	38.86	38.96		
FeO	17.75	17.78	17.81	17.50	17.48		
MgO	41.96	42.06	42.08	42.19	41.87		
CaO	0.26	0.27	0.25	0.26	0.28		
Cr ₂ O ₃	0.31	0.29	0.32	0.29	0.31		
MnO	0.19	0.20	0.18	0.19	0.19		
NiO	0.02	0.02	0.04	0.02	0.05		
	99.55	99.46	99.67	99.31	99.14		

Small crystals							Dendrite
	74001,2-2	74001,2-11A	74001,2-11B	74001,2-14	74001,2-15	74001,2-16	74001,2-9
SiO ₂	38.10	38.62	38.34	38.54	38.22	38.09	38.31
FeO	20.97	20.10	20.62	19.71	20.06	20.34	21.07
MgO	39.31	39.66	39.15	40.09	40.15	39.96	39.01
CaO	0.31	0.33	0.35	0.32	0.30	0.34	0.33
Cr ₂ O ₃	0.35	0.38	0.40	0.37	0.32	0.35	0.39
MnO	0.24	0.24	0.20	0.22	0.21	0.22	0.21
NiO	0.05	0.05	0.04	0.04	0.04	0.04	0.04
	99.33	99.38	99.10	99.29	99.30	99.34	99.36

fraction) and for 74220,66A (90–150 μm fraction)] by REID *et al.* (1973). Some of the droplets in 74001 contain ovoid vesicles which in some cases are open to the exterior (Fig. 7c). In the 250–500 μm size fraction, 2 per cent of the unbroken droplets contain single vesicles; and in the 45–75 μm size fraction 8 per cent of the unbroken droplets contain single vesicles. The droplets containing vesicles are usually simple rather than composite and are in most cases either partly glassy or very finely crystallized. Vesicles in the coarsely crystallized, barred droplets are rare.

B. Chemistry

The petrographic homogeneity of the orange glass spheres is supported by a distinctive chemical homogeneity (Table 2). Compared to most lunar glasses, the orange glass has high Ti, Fe, Mg and low Si, Al, Na and K (REID *et al.*, 1973a; GLASS 1973). Analyses of individual droplets are very close to the analysis of the bulk soil (RHODES, 1973) (Table 2); the normative composition corresponds to an ilmenite pyroxenite with 17 per cent ilmenite, 22 per cent olivine, 41 per cent pyroxene and 18 per cent feldspar (REID *et al.*, 1973). The compositions of glassy droplets from sample 74001, crystallized droplets, and remnant glass in crystallized droplets (Table 3) are similar and are close to the composition of the orange glass. The major element composition of the orange glass can be approximated by adding olivine to the mare lava composition (REID *et al.*, 1973a). However, trace element composition of the orange glass differs considerably for some elements from that of the Apollo 17

Table 2. Chemical analyses of the Apollo 17 orange soil (74220)

	REID <i>et al.</i> (1973a)		PRINZ <i>et al.</i> (1973)	REID <i>et al.</i> (1973b)		CARTER <i>et al.</i> (1973)	GLASS (1973)	RHODES (1973)	ROEDDER and WEIBLEN (1973)	RHODES (1973)
	Orange glass	Varia- bility	Orange soil	Orange glass in 74240	Varia- bility	Orange glass	Orange glass	Bulk Anal. orange soil 74220	Orange glass	17- 'Subfloor' basalt
SiO ₂	38.55	(0.26)	39.2	38.63	(0.33)	38.0	39.4	38.57	39.5	38.77
TiO ₂	8.87	(0.13)	9.4	8.96	(0.47)	8.87	9.3	8.81	8.56	12.04
Al ₂ O ₃	5.85	(0.11)	5.8	5.87	(0.36)	5.51	6.02	6.32	6.41	9.09
Cr ₂ O ₃	0.55	(0.01)	0.61	0.55	(0.3)	0.70	—	0.75	0.52	0.43
FeO	21.96	(0.27)	22.4	22.0	(0.32)	22.4	22.7	22.04	22.2	18.77
MgO	14.99	(0.44)	14.1	14.79	(1.15)	14.5	15.5	14.44	14.4	8.42
CaO	7.16	(0.14)	7.6	7.31	(0.40)	6.99	6.42	7.68	7.13	10.93
Na ₂ O	0.33	(0.02)	0.31	0.37	(0.13)	0.39	0.69	0.36	0.51	0.37
K ₂ O	—	—	0.04	0.08	(0.02)	0.06	0.0	0.09	0.08	0.06
P ₂ O ₅	—	—	0.04	—	—	—	—	0.04	0.05	0.07
MnO	—	—	—	—	—	—	0.30	0.30	0.32	0.28
S	—	—	—	—	—	—	—	0.07	—	0.17
No. of samples	47	—	?	80	16	—	19	1	?	1

Table 3. Chemical analyses of glasses from 74001,2

	95% crystallized 74001,2-4	Clear orange glass 74001,2-5	Surface droplets 74001,2-6	Surface droplets 74001,2-7	Clear orange glass 74001,2-8	Average per all grains	Variability
SiO ₂	39.38	38.38	39.14	38.61	38.16	38.73	1.22
TiO ₂	10.34	9.19	9.03	9.81	8.95	9.46	1.39
Al ₂ O ₃	6.06	5.91	5.82	6.33	5.78	5.98	0.55
Cr ₂ O ₃	0.56	0.59	0.54	0.71	0.56	0.59	0.17
FeO	22.44	22.25	23.29	22.84	22.16	22.60	1.13
MgO	10.29	14.87	14.54	16.16	15.12	14.20	5.87
CaO	9.55	7.40	7.32	7.08	7.29	7.73	2.47
Na ₂ O	0.32	0.40	0.37	0.25	0.40	0.35	0.15
K ₂ O	0.06	0.07	0.07	0.06	0.08	0.07	0.01
	99.00	99.06	100.12	101.85	98.50	99.71	

high-Ti basalts and the trace element content of the orange glass cannot be made by simply adding olivine to basalt (RHODES, 1973). For example, Sr is higher in the orange soil compared to Apollo 17 mare basalt (RHODES, 1973) and volatile elements such as Zn, Cu, Ga, Pb, Te, Br, Cl, and F are also highly enriched in the orange soil compared to mare basalts or normal soils (RHODES, 1973; WANKE *et al.*, 1973; JOVANOVIĆ *et al.*, 1973, TATSUMOTO *et al.*, 1973). Much of the Cl and Br in the orange soil is easily leached with hot water (JOVANOVIĆ *et al.*, 1973) which may indicate a surface site for these elements. Similarly, 80 per cent of the Pb is leached from the orange soil by 1.5 N HCl, again suggesting a surface correlation (TATSUMOTO *et al.*, 1973). Volatilization studies by SILVER (1974) reveal that 60–80 per cent of the total lead is released below 600°C. By drilling through the exterior surface of orange spheres with the ion microprobe, C. Meyer (Presentation, Fifth Lunar Science Conference) determined that lead is extremely concentrated on the outermost surface but is below detection limits in the interior. Meyer proposes that

this surface lead may be present in micromounds. The orange and black soil is also relatively rich in surface correlated gases; H_2O , CO_2 , CO and SO_2 are released from both 74220 and 74001 at relatively low temperatures, less than 400–450°C (GIBSON and MOORE, 1973a, WSZOLEK *et al.*, 1973). EPSTEIN and TAYLOR (1973), however, report very low values of total carbon and H_2O . GIBSON and MOORE (1973) discuss the variability of the content of these low temperature released glasses among different splits of 74220 and suggest that this variability, particularly for CO_2 and CO , may in part, be associated with the color differences and zoning in the original orange soil aggregate.

C. Surface features

The abundance of easily leachable species and gases released at low temperature in the orange and black glass is almost certainly indicative of a site for these species and a case can be made for a vapor condensation mechanism which originally concentrated these species on the surface of the glass. We have examined the surfaces of many of the droplets from 74220 and 74001 with the SEM. Many of these surfaces display micromounds which range from 200 Å to more than 1000 Å in diameter (Fig. 8). In many places, the tightly packed mounds are scraped away or piled up by low velocity impacts and abrasion. Similar micromounds are found on Apollo 15 green glass spheres and are interpreted as a vapor condensate (MCKAY *et al.*, 1973). Some features of the surface morphology of the orange and black droplets are similar to features found on glass droplets formed in Hawaiian lava fountain eruptions. These features include adhering mounds and droplets (Fig. 8b) of glass from low velocity collisions. Micromounds are also present on many of the Hawaiian spherules (Fig. 8b) and some show abrasion and scrape marks apparently formed in the fire fountain plume.

D. Lunar surface exposure age

The orange and black droplet samples contain only a trace of agglutinate grains; this implies that there has been very little exposure of these soils to the lunar surface environment (MCKAY *et al.*, 1970, 1971; DUKE *et al.*, 1970). The friable 'clod' consists of 100 per cent orange glass. None of the surfaces of individual droplets studied were pitted by high velocity micrometeorite impact craters, another indicator of lunar surface exposure (MCKAY and HEIKEN, 1973). Cosmogenic gas exposure age is ~ 30 m.y. (KIRSTEN *et al.*, 1973; HUNEKE *et al.*, 1973). KIRSTEN *et al.* infer that the orange glass deposit near the lunar surface was covered within 30 m.y. of its deposition by several meters of material and then excavated less than 30 m.y. ago by the Shorty Crater impact. EBERHARDT *et al.* (1973) have determined an $^{38}\text{Ar}_{\text{sp}}/^{37}\text{Ar}$ ratio which gives an exposure age of 40 m.y. (average of four size fractions) for sample 74220. SCHAEFFER and HUSAIN (1973) agree with this model, stating that the short exposure age indicates that the deposit was buried to a depth of at least several meters most of the time since its formation. It is possible that the original deposits of glass droplets were rapidly buried by later basalt flows or a large impact blanket, protecting them from the solar wind and micrometeorites until excavation by later large impact craters; this speculation is compatible with the distribution of abundant basalt fragments on the rim of Shorty and other craters in the area.

STORZER *et al.* (1973) have determined a solar flare track age of about 10 m.y. for the orange glass; this age may be closer to the actual age of Shorty Crater.

E. Glass formation age

An $^{40}\text{Ar}/^{39}\text{Ar}$ gas retention age of 3.54 ± 0.05 b.y. has been determined by HUNEKE *et al.* (1973) who conclude that this age most likely represents the actual formation age of the orange glass. HINTENBERGER and WEBER (1973) also determine an argon age of 3.5 ± 0.3 b.y. for the orange glass. SCHAEFFER and HUSAIN (1973) determine a somewhat older glass formation age of 3.71 ± 0.06 b.y. again using the $^{40}\text{Ar}/^{39}\text{Ar}$ technique. TATSUMOTO *et al.* (1973) assuming a two stage U-Pb evolution, calculate a formation age of 3.63 b.y. for the orange glass. Using three stage Pb evolution models, an age as high as 3.83 b.y. can be calculated (P. D. Nunes, written communication). So, depending on the techniques and the assumptions used, the formation age of the orange glass may be as young as about 3.5 b.y. or as old as 3.83 b.y. This age can be compared to the formation age of the mare basalts at the Apollo 17 site. Mare basalt samples collected at the Apollo 17 landing site have ages of 3.82 ± 0.06 AE (Rb-Sr, EVENSEN *et al.*, 1973), 3.82 ± 0.5 AE ($\text{Ar}^{39}/\text{Ar}^{40}$, KIRSTEN *et al.*, 1973), 3.71 ± 0.06 AE ($\text{Ar}^{39}/\text{Ar}^{40}$, SCHAEFFER *et al.*, 1973), and 3.83 ± 0.10 AE (Rb-Sr for sample 75055, TATSUMOTO *et al.*, 1973).

It is apparent that the orange glass is, to a first approximation, the same age as the mare basalts. As outlined above, the low total exposure age of the orange glass require that this deposit was rapidly buried to a depth of at least a few meters. Based on the abundance of basalt around Shorty Crater and the predominance of basalt components in normal dark mantle soils and, we infer that the capping material was a basalt flow rather than impact ejecta (regolith). As outlined by APOLLO 17 P.E.T. (1973), HEIKEN and MCKAY (1973), and earlier in this paper, the regolith consists of primarily comminuted basalt. There are, however, orange and black spheres present in nearly all of the Apollo 17 mare soils. In this model, the orange and black glass deposits are interbedded with basalt flows; this is in agreement with the approximately concordant ages of orange glass and basalt.

ORIGIN OF THE ORANGE AND BLACK GLASS

It is extremely important to understand the origin of the orange and black glass. If this material is impact produced, its unusual properties may reveal important new information on impact processes and possibly on the composition of the impacting body. On the other hand, if the orange and black glass was produced by endogenous processes, the unusual composition of this material may reveal considerable information about the lunar interior and distribution of these deposits may be a valuable clue to the history of mare volcanism. We consider below several possible hypotheses and discuss the relative merits of each.

A. Lava fountain hypothesis

Origin of the orange glass in a lava fountain eruption was proposed in the report of the Preliminary Examination Team (Apollo 17 P.E.T., 1973) and by CARTER *et al.* (1973), MCKAY and HEIKEN (1973) and REID *et al.* (1973a). A similar model was proposed for the formation of the Apollo 15 green glass droplets which have textural and morphological characteristics very similar to those of the orange and black glass (MCKAY *et al.*, 1973; AGRELL *et al.*, 1973).

It is known that petrographically and chemically homogeneous, vesicular glass droplets are formed in lava fountains of terrestrial basaltic magmas (HODGE and WRIGHT, 1964; AGRELL *et al.*, 1970; HEIKEN, 1972). Microprobe analyses of 40 droplets formed in lava fountains at Mauna Ulu, Hawaii indicate very little variation between glasses (REID *et al.*, 1973a). The homogeneity of the glasses appears to be characteristic of vitric volcanic ash from a single eruption. The glass in the largest obsidian flow at Newberry Crater, Oregon is also remarkably chemically homogeneous (LAIDLEY and MCKAY, 1971). In contrast, glasses produced in terrestrial and lunar impacts generally exhibit petrographic and chemical heterogeneity; they often contain some shock metamorphosed detritus and in some cases, distinct bands of glass with a different refractive index (CHAO, 1967; HEIKEN, 1972). Tektites, possibly formed by impact, also are characterized by flow structure and lechatelierite or other glass inclusions (CHAO, 1968; GLASS, 1968). These observations suggest to us that the orange glass droplets were formed from a mafic liquid in a lava fountain.

The composite glass droplets of the orange and black glass at the Apollo 17 site, green glass at the Apollo 15 site, and orange glass at the Apollo 11 site appear to have been impacted at *low velocity* by molten glass and solid particles of the same chemical composition. The resulting grain is lumpy in appearance, with nearly spherical to plate-like forms welded to the grain surface. Low velocity spalls and pits mark the surface of some grains (Fig. 8c); these probably resulted from collisions with other grains or by falling to the lunar surface. Most of the droplets were solid before reaching the lunar surface; the orange glass cooled at a rate of 400–500°C/min (UHLMANN *et al.*, 1973) and typical calculated flight time of ejecta in a typical lunar cinder cone is 40–80 sec (MCGETCHIN and HEAD, 1973). This flight time is sufficient for most of the droplets to become relatively hard before falling to the lunar surface. By comparison, nearly half of the droplets formed in a lava fountain at Mauna Ulu, Hawaii (1972 eruption) are spalled or fractured (Table 4). Surface morphology of some Hawaiian spherules suggests that part of the fracturing was caused by collisions with other droplets.

If the orange glass droplets were formed in a nearly vertical jet or lava fountain, then composite droplets may have formed by falling back through the fountain, and colliding with molten or semi-molten droplets. In some cases, composite droplets may have been again thrown up by such collisions. Slower cooling or reheating associated with fallback through the jet or recycling may have triggered crystal

Table 4. Comparison of grain shapes; orange glass and Hawaiian tephra

	% 74220,85 (orange glass only)				1959 Kilauea Iki	1959 Kilauea Iki	Mar. 1972 Mauna Ulu	Mar. 1972 Mauna Ulu
	A	B	C	D	<1 mm	>1 mm	>1 mm	<1 mm
Spherical	20.5	29.8	20.2	31.7	1.4	5.8	8.7	16.4
Ovoid	4.5	11.4	4.6	2.4	3.1	7.4	47.2	49.8
Pele's Hair	0.3	—	—	—	12.8	4.9	0.5	7.6
Broken droplets	50.5	42.6	62.4	52.8	63.8	78.5	40.0	21.2
Angular glass frag.	24.1	16.1	12.8	13.0	18.8	3.6	3.5	5.0
Total no. grains	307	298	109	123	484	554	540	420

growth, resulting in the correlation between composite droplets and crystallization in the glass (REID *et al.*, 1973; MCKAY and HEIKEN, 1973). Droplets leaving the plume at lower angles are exposed more quickly to a cold environment and can quench to a clear glass. Droplet textures ranging from homogeneous glass to spherulites and dendrites may reflect a decreasing degree of supercooling. The relationship between spherulitic and dendritic texture and decreasing degrees of supercooling has been determined experimentally by LOFGREN (1974) and is in accord with the proposed lava fountain mechanism for producing these textures in composite droplets.

Studies made of terrestrial volcanic eruptions and lava fountains show that high ejection angles are typical. A ballistic study of Northeast Crater, Etna, Italy, by McGETCHIN and HEAD (1973) and McGETCHIN *et al.* (in press) reveals that the ejection angles of bombs and droplets ranged from 70° to vertical (with a median of 84.3°) and that the median velocity of particles within the jet was 51 m/sec.

A very detailed study of two Strombolian eruptions by CHOUET (1973) indicated that the median dispersion of particles from the jet axis was 8° for eruption 1 and variable for eruption 2. The average velocities were 26.2 m/sec for eruption 1 and 15.3 m/sec for eruption 2. A study of photographs of Hawaiian lava fountains indicates the ejection angles ranged from 60° to vertical, except when a blocked conduit resulted in a deflected, low angle jet.

The dearth of vesicles in the orange glass droplets appears to argue against a fire fountain origin; terrestrial lava fountains appear to be driven mostly by a release of gases as the magma reaches the Earth's surface. However, if the lunar lavas had a very low viscosity, it is possible that much of the gas escaped before the droplets cooled. Calculated viscosities, using the method of SHAW (1972) indicates that the orange glass had a viscosity of 3.5 poise at 1250°C. The actual value of about 10 poise was experimentally determined by CUKIERMAN *et al.* (1973). For comparative purposes, the calculated viscosity of droplets from Mauna Ulu, Hawaii lava fountain is 145 poise at 1250°C. This suggests that if a gas phase was present it might have escaped from very low viscosity fluids into the lunar vacuum before solidification. The vesicles present in the finely crystallized droplets in 74001 may indicate that an ambient gas phase was present in the fire fountain plume which created enough external pressure to prevent complete degassing before solidification. The presence of vesicles in the finely crystallized droplets eliminates one of the main objections to the fire fountain hypothesis by ROEDDER and WEIBLEN (1973).

Pele's hair commonly forms in terrestrial fire fountains but small droplets are not uncommon. The order of magnitude difference in viscosity may account for the lack of Pele's hair and the abundance of droplets in lunar fire fountains; streamers of more fluid magma may break up into droplets much more readily (ISARD, 1971).

The fire fountain hypothesis can be used to explain many of the other features of orange and black droplets. The olivine phenocrysts may represent true volcanic phenocrysts present in the magma before eruption. The concentration in surface correlated volatile elements and surface bound gas species may represent condensation of the volatile elements in the fire fountain and may reflect the presence of an ambient gas phase. The micromounds observed on these droplets are strong evidence for the reality of this condensation process; these micromounds may contain trapped volatile metals.

The relatively oxidized state of the orange glass compared to most lunar material may also be associated with a relatively oxidizing gas phase with high CO_2/CO ratio compared to other lunar material. MAO *et al.* (1973) did not detect Fe^{3+} in the orange glass which means that if present it must be less than 1 wt. %, their limit of detection by Mossbauer techniques. Experiments with synthetic glasses by MAO *et al.* (1973) provide a rough bracket for $p\text{O}_2$ of the orange glass of $10^{-8.1}$ and $10^{-9.1}$.

Origin of the orange and black glass as a fire fountain deposit is also in agreement with formation ages and the inferred stratigraphy which suggests that the glass deposits are adjacent to or interbedded with basalt flows and must therefore be roughly contemporaneous with the volcanic activity which produced the Apollo 17 basalts. HUNEKE *et al.* (1973) and SCHAEFFER and HUSAIN (1973) argue that the age correlation between the orange glass and the Apollo 17 mare basalts is strong evidence that the two are related and that the orange glass is volcanic.

B. Impact hypothesis

Most of the glass particles found in lunar soils have been formed by impact melting of lunar rock or regolith. Consequently it is reasonable to consider the possibility that the orange and black glass may also have been formed by impact.

Most of the glass in lunar soils is either heterogeneous or contains shocked crystalline debris. The abundance of extremely homogeneous glass droplets rules out the possibility that a small impact formed the orange and black glass. A large impact, however, might completely melt and homogenize large volumes of material. The composition of the orange glass requires that such an impact occur in a mafic cumulate (REID *et al.*, 1973). The impact which originally formed the mare basins might be sufficiently large to create and homogenize large volumes of glass of cumulate composition. However, the age of the glass is too young to be related to the large basin impacts and the inferred stratigraphic position of the glass deposits also precludes this source. Also, the presence of preexisting phenocrysts in the glass droplets argues against formation of the glass by impact. Finally, the lack of any rock fragments of similar composition is difficult to explain with the impact hypothesis. No convincing model for impact generation of orange droplets from solid rock has been advanced which answers these objections.

C. Lava lake impact hypothesis

An interesting variation of the impact hypothesis proposed by ROEDDER and WEIBLEN (1973) overcomes some of the objections to the simple impact hypothesis. In this variation the meteorite impact occurred in a lava lake rather than in crystalline rock. Large volumes of molten magma were already present, possibly with olivine phenocrysts, and this magma was splashed out by the impact forming the deposits of glass droplets. This hypothesis, however, does not explain many of the features associated with the orange glass such as the enrichment in volatile metals, the enrichment in surface correlated gas released at low temperatures, and the relatively more oxidized state. Furthermore, ejecta from impact into a lava lake would be expected to include crystalline basalt fragments of orange glass composition from the solid crust which must rapidly form on lava lakes. After complete cooling of the lava lake, subsequent impacts should also produce crystalline ejecta; no basalt fragments

of orange glass composition have been reported. PRINZ *et al.* (1973) propose that the large impacts which formed the mare basins triggered rapid melting which formed an ultramafic lava lake. The lava lake was then splashed by other impacts or developed lava fountains to produce the orange and green glass droplets. At some later time, less mafic basalt flows covered the cooled lava lake.

The strongest argument against this hypothesis arises from the ages of the orange and green glasses which are both as young as or younger than the youngest basalt at their respective sites. The green glasses are 3.38 b.y. old which is about 600 m.y. younger than the Imbrium basin excavation time. It is unlikely that the two are related (PODOSEK and HUNEKE, in press; HUNEKE *et al.*, 1973).

D. Vapor condensation hypothesis

A large impact creates a cloud of vaporized rock and some of the vapor could condense into droplets. However, it is highly unlikely that the orange glass droplets could have condensed from a siliceous vapor; the presence of olivine phenocrysts, vesicles, and low velocity impact features cannot be reconciled with an origin by vapor condensation.

LOCATION OF ORANGE AND BLACK GLASS DEPOSITS

Glass of composition very similar to the Apollo 17 orange glass has been found in material from each of the Apollo sites (e.g. MAO *et al.*, 1973, KEIL *et al.*, 1970). In particular WOOD *et al.* (1970a, b) found in the Apollo 11 coarse fines a breccia or clod consisting mainly of orange glass spheres and fragments in a matrix of fine-grained glass. The composition of this glass is nearly identical to that of the Apollo 17 orange glass. They also describe a clod of similar texture from 12033 coarse fines, but the glass in this clod has a uniquely high TiO_2 (16 per cent) composition (WOOD *et al.*, 1971). The most abundant homogeneous glass deposits occur at the Apollo 11, 15 and 17 sites. These landing sites are all located near the faulted rims of large ring basins. The ring basins, now filled with mare lavas, appear to have been formed by the impact of planetesimals into the lunar surface. The fractured zones around the rims appear to have acted as conduits for later volcanism. The source for extensive Imbrium lava flows, which cover an area of $2 \times 10^5 \text{ km}^2$, is a 20 km long fissure located at $22^\circ 50' \text{N}$ and $31^\circ 20' \text{W}$ (southwest rim of the Imbrium basin). The source area is a low albedo cinder cone complex similar in appearance to some of the dark mantle deposits along the south rim of Mare Serenitatis (SCHABER, 1973). This fissure is located at the junction of ring faults of the Imbrium and Copernican basins (the latter is an old mare basin obscured by younger deposits). The same structural relationship exists at the Apollo 17 site (Serenitatis and Tranquillitatis basins), the Apollo 15 site (Serenitatis and Imbrium basins) and the Apollo 11 site (Tranquillitatis and Nectaris basins). If eruptive vents are located at ring basin junctures, as observed by SCHABER (1973), then it is possible that vents are located near the Apollo 11, 15 and 17 landing sites.

Newberry Caldera, Oregon, is one of many examples of volcanoes located at junctions of fault zones. Newberry Caldera is a major shield volcano located at the junction of the Brothers fault zone and the Walker Rim fault zone (HIGGINS, 1973). On Earth, lava fountaining is a local phenomenon, rarely depositing ash more than

a few km from the vent area (unless a strong wind affects the distribution). Even in lunar conditions, ash deposits from lava fountains should be local, extending no more than tens of kilometers beyond the vent. This is in accordance with the location of sites at the basin edges, where homogeneous glass droplets were sampled.

PIETERS *et al.* (1974), using laboratory and telescopic reflectivity spectra, conclude that the soils derived from basalts at the Apollo 17 landing site are similar to the Serenitatis dark interior ring and to the surface of much of mare Tranquillitatis. Northwest of the Apollo 17 landing site, south of Crater Le Monnier, the relative spectral reflectivity of the dark mantle superimposed on the lavas is similar to sample 74001 (black droplets) which the authors call 'pure' dark mantling material.

The dark mantled deposits around the inner rim of mare Serenitatis, west of the Apollo 17 landing site, may truly be partly of pyroclastic origin. They may, however, be nearly contemporaneous with the mare lavas and not young as was suggested in pre-mission analyses of the dark mantle.

In the normal soil samples from the Apollo 17 dark mantle site, the small proportion of black droplets may contribute to the low albedo, but as the bulk of this soil is made of high Ti basalt fragments, agglutinates, and related high Ti glass forms, the low albedo must be primarily the result of this material and not the black droplets.

The crew of Apollo 17 observed that orange soils may be interbedded with lavas and dark mantle in southern and southeastern mare Serenitatis between Taquet crater and the Sulpicius Gallus Rilles: "This whole dark mantle in here around Sulpicius Gallus . . . there are scattered craters with a variety of orange to red-brown hues . . . and they all, except for that large rimless depression, which looked as if it was exposing some layers which were streaming those colors—that colored debris down its walls—all the other craters seem to be small impacts that apparently are penetrating just far enough into dark mantle material to tap this zone of orange to red-brown material." (LMP Schmitt, transcript, Apollo 17 mission.) LUCCHITTA and SCHMITT (1974), based on visual observations and flight film, show that orange material in the Sulpicius Gallus region is found as haloes, patches, or rays around 50 to 250 m diameter fresh craters; all 'orange halo' craters are located in patches of very dark material. They believe that the orange layers may occur in the subsurface as continuous layers or possibly discrete pockets to depths of about 50 m. The Sulpicius Gallus area is nearly 400 km from the Apollo 17 site but is also near the margin of Serenitatis, lending support to the concept that the orange glass represents a pyroclastic ejecta from vents around the basin rim.

SUMMARY AND CONCLUSIONS

(1) Orange glass droplets from the Apollo 17 site are both chemically and texturally homogeneous; none that were investigated contain shock damaged crystals which are common in glass produced by meteorite impacts. Black droplets are apparently the partially crystallized equivalents of the orange glass.

(2) Chemically and texturally homogeneous glass droplets are known to form in terrestrial lava fountains of basaltic melt. By analogy, orange glasses from the Apollo 11 and 17 sites and green glasses from the Apollo 15 site may have formed in lava fountains of low viscosity lunar basaltic magmas.

(3) The regolith at the Apollo 17 site contains an average of about 5 per cent orange and black droplets. They may be derived from a layer or layers located 10–30 m below the valley floor. It is possible that the orange and black droplet layer or layers may be interbedded with mare lava flows. Deposits of orange and dark gray soils were also observed in ejecta from impact craters around the southern rim of mare Serenitatis. Depending on techniques and assumptions, calculation formation ages of the orange droplets range from 3.5 to 3.83 b.y.; mare lava samples at the same site have ages of $3.71\text{--}3.83 \pm 0.1$ b.y. This age relationship suggests a possible relationship between the deposits of homogeneous glass spheres and dated mare lava flows, namely, that they may be part of the same volcanic episode although not the same eruptions.

(4) The homogeneous glass droplets sampled at the Apollo 11, 15 and 17 sites are located on or near mare basin rims. Dark mantle deposits, interpreted as pyroclastic blankets in some locations, and numerous rilles are also present on the mare surface, near basin rims. The orange and green glass droplets may be concentrated locally near such vent area. This association is in accordance with the limited extent of ash deposits from terrestrial lava fountains (< 10 km from the vent).

(5) The dark mantle deposits of McGETCHIN and HEAD (1973) and LUCCHITTA (1973) may indeed have a pyroclastic origin. The Apollo 17 results suggest that they are, however, not young, but associated with mare volcanism.

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